

ECON 21030: Honors Econometrics

Homework 4

Spring Quarter 2024

Question 1: [40 points] Government agencies and private foundations distribute billions of dollars in financial aid to college-goers each year, in the hopes of boosting educational attainment. Tennessee, for example, guarantees no-cost tuition for all high school graduates at in-state community and technical colleges. In this exercise, you'll investigate the potential impact of this state-run program on Tennessee college enrollment.

1. The first step is to assemble data for the set of Tennessee colleges over 2011-2016 from the Integrated Postsecondary Education Data System (IPEDS). These data are publicly available through the U.S. Department of Education.

To download the data files, visit the IPEDS [homepage](#), click “Use the Data” and then select “complete data files” from the dropdown menu in the “Survey Data” section. You'll draw on two sections of the IPEDS surveys:

- “Directory Information” (DI) files from the “Institutional Characteristics” survey
- “Student Financial Aid and Net Price” (SFA) files from the survey by the same name

Please download files listed under the “Data Files” category (as opposed to the “Stata Data Files” category). You should not use the revised data files. IPEDS identifies these files with the suffix “_rv”. Likewise, do not use any of the revised or imputed variables in the dataset. The original reported values will suffice.

Your next goal is to combine the files into one large dataset. You can do this in a few ways, but here is some free advice:

- Create a loop that reads in each dataset, generates an identifying year variable for this dataset, drops any variables you do not need for the analysis (see below), and saves a temporary dataset for each year and file type (DI and SFA).
 - Append together (ideally also in a loop) the temporary datasets across years to construct two panel data sets— one for DI and one for SFA. The rows of each dataset should be uniquely identified by “unitid” and “year.”
 - Merge together the DI and SFA panel data sets to create a single panel dataset with each row uniquely identified by “unitid” and “year.” Keep only rows that are found in both the DI and SFA panels.
2. The next step is to clean the panel dataset you've created. Considering our research question, you should make a few sample restrictions:
 - Drop institutions not in Tennessee

- Restrict to active institutions (see `cyactive` in the dictionary file)
- Restrict to 2 and 4 year institutions (see `iclevel` codes in the [codebook](#))

Your cleaned analysis data should contain the following variables:

- `ID_IPEDS`: a unique identifier for each institution
- `year`: the four-digit academic year; e.g. 2010-11 should be coded as 2010
- `degree_bach`: a dummy variable that identifies bachelor's degree-granting institutions (`degree_bach=1` if 4-year, 0 otherwise; again see `iclevel` codes in the [codebook](#))
- `public`: a dummy variable what identifies public institutions (`public=1` if public, 0 otherwise; see control codes in the [codebook](#))
- `enroll_ftug`: the total number of first-time, full-time undergraduates in each institution and year
- `grant_state`: total amount of state and local grant aid awarded to first-time, full-time undergraduates
- `grant_federal`: total amount of federal grant aid awarded to first-time, full-time undergraduates

Your cleaned analysis data should satisfy the following criteria:

- The data should cover academic years 2010-11 through 2015-16. To construct a balanced panel, you should include only the colleges in Tennessee for which data exist in all years
- The sample should contain only undergraduate institutions, which may offer bachelor's and associate degrees as well as diplomas and certificates. Drop institutions that only offer graduate degrees
- The sample should include both public and private institutions
- You should use total count variables where they exist, rather than manually adding subcategories
- You should drop any institutions that change `degree_batch` or `public` over the included years

You can verify that your data have been cleaned correctly by computing summary statistics of `year`, `ID_IPEDS`, `degree_bach`, `public`, `enroll_ftug`, `grant_state`, and `grant_federal`. They should look very similar (if not identical) to this table.

stats	year	ID_IPEDS	degree_bach	public	enroll_ftug	grant_state	grant_federal
mean	2012.5	259717.7	.4907407	.4351852	548.0278	1201195	1294725
sd	1.709144	81429.63	.5003004	.4961642	706.7017	2582446	1560033
N	648	648	648	648	648	648	648

3. We're now ready to analyze the cleaned data. The Tennessee Promise guarantees no-cost tuition for all Tennessee high school graduates who attend the state's public community and technical colleges. For the sake of concision, we'll refer to schools that offer bachelor's

degrees as “four-year colleges” and schools that don’t as “two-year colleges” (we recognize that, in practice, time to degree completion varies widely across students and credentials). We’ll also distinguish between public and private schools, so your analysis will compare four groups:

- Public, two-year colleges
- Public, four-year colleges
- Private, two-year colleges
- Private, four-year colleges

Given these categories, community and technical colleges will fall in the “public, two-year” group. Tennessee Promise scholarships were first offered to students in the 2015-16 academic year. To start your analysis, create a graph that compares average school-level state plus local grant aid across the four types of institutions during the sample period. Describe the results.

4. Specify a regression model of your choice that you believe will generate the most accurate estimate of the causal effect of the Tennessee Promise program on enrollment at public, two-year colleges. Estimate this model and present your findings in a table.
5. Discuss the key assumptions required for the causal interpretation of your estimates, as well as any concerns you might have with using your results to make a recommendation to the Illinois State Board of Education about whether or not to implement a similar program.
6. Now keep only the data for two-year colleges in 2014 and 2015. Estimate this regression:

$$Y_{it} = \alpha + \gamma Public_i + \tau Post_t + \beta D_{it} + \epsilon_{it}$$

where $Public_i \in \{0, 1\}$ indicates a public college, $Post_t \in \{0, 1\}$ indicates 2015, and $D_{it} = Public_i \cdot Post_t$.

Report in a table your estimate of β with both a heteroskedasticity-robust standard error and a standard error that clusters by college. How do these standard errors compare, and which would you rather use for inference on β ?

7. Duplicate your dataset, generating a new college ID which takes on a different value for the same college in the two copies. Estimate the same regression and report in the same table as in (a) your estimate of β with both a heteroskedasticity-robust standard error and a standard error that clusters by the new college ID. Compute the ratio of standard errors on D_{it} here relative to those you obtained in (a). Are you surprised by the answer?
8. Estimate the same regression in the duplicated dataset but now cluster the standard errors by the original college ID. Compare this standard error to the one you obtained in part 8. You do not need to add anything to your regression table for this question.
9. Return to the original, non-duplicated dataset from part 8. Run a simulation, similar to that from the Bertrand et al. (2004) paper we discussed in class, which randomly re-assigns $Public_i$ across colleges (holding Y_{it} fixed). Re-estimate the regression in each of these “placebo samples,” with both heteroskedasticity-robust and clustered standard errors, and record whether you can reject the null hypothesis of $\beta = 0$ at the 1%, 5%, and 10%

confidence level in each. Repeat this simulation 500 times and report in a new table the average rejection rates for the two standard error estimators and three confidence levels. What do you conclude?

10. Transform the non-duplicated dataset to be in first-differences: construct a new dataset where each row corresponds to a college and one of the variables is the difference in Y_{it} between 2014 and 2015 (call this ΔY_i). Estimate the bivariate regression of ΔY_i on $Public_i$ with heteroskedasticity-robust standard errors and report this in the table from above. Discuss how this estimate and standard error compares with those in estimated in the non-first differenced dataset.
11. Repeat your simulation exercise on this first-differenced specification, using heteroskedasticity-robust standard errors. Report the average rejection rates in the same table as in (d). Discuss your findings: why do you get different answers here than you did before with heteroskedasticity-robust standard errors?